

Numerical Approximation of the Spectrum for the Dirichlet Laplacian

Ebube Anozie

APPM 6470 – Prof. Nancy Rodriguez

April 24, 2026

Outline

Numerical
Approximation of
the Spectrum

Ebube Anozie

Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion

1 Introduction and Motivation

2 Finite Element Discretization

3 Convergence Theory

4 Numerical Experiment

5 Conclusion

The drum analogy

Numerical
Approximation of
the Spectrum

Ebube Anozie

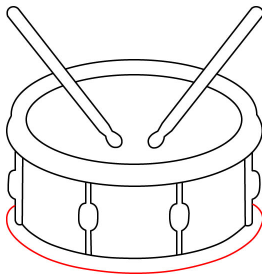
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion



- When we see a drum, we have an idea of how it would sound.
- Geometric information $\implies$ spectral information.

The Dirichlet Laplacian

Numerical
Approximation of
the Spectrum

Ebube Anozie

Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion

We consider the eigenvalue problem:

Find eigenpairs (λ, u) :

$$\begin{aligned} -\Delta u &= \lambda u && \text{in } \Omega, \\ u &= 0 && \text{on } \partial\Omega. \end{aligned}$$

- Exact eigenvalues known only for simple shapes
- Numerical methods needed for general domains

Weak form

Find $(\lambda, u) \in \mathbb{R} \times H_0^1(\Omega)$, $u \neq 0$:

$$a(u, v) = \lambda b(u, v) \quad \forall v \in H_0^1(\Omega),$$

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx$$

$$b(u, v) = \int_{\Omega} uv \, dx$$

How can we turn an abstract, infinite-dimensional eigenvalue problem into something a computer can solve?

Galerkin idea

Infinite-dimensional problem $\rightarrow$ finite-dimensional matrix problem

- Restrict to a finite-dimensional subspace $V_h \subset H_0^1(\Omega)$.
- Choose basis $\{\phi_j\}_{j=1}^N \Rightarrow u_h = \sum_j u_j \phi_j$.
- Solve

$$A_h \mathbf{u} = \lambda_h M_h \mathbf{u},$$
$$(A_h)_{ij} = a(\phi_j, \phi_i), \quad (M_h)_{ij} = b(\phi_j, \phi_i).$$

The Galerkin "gamble"

Numerical
Approximation of
the Spectrum

Ebube Anozie

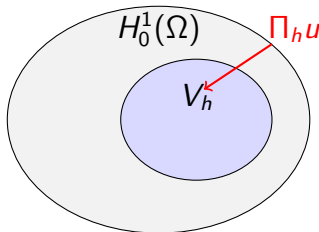
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

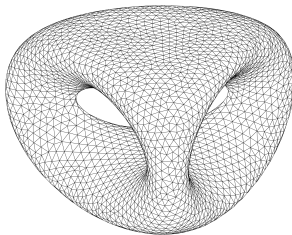
Conclusion



If $\{V_h\}$ is *approximating* (dense in H_0^1 as $h \rightarrow 0$), then the finite-element eigenvalues converge to the exact eigenvalues.

P1 Finite Elements

- Cover the domain Ω with a triangle mesh $\mathcal{T}_h$.
- $h =$ maximum element diameter
- Approximate space V_h :
 - functions are continuous, linear on each triangle,
 - vanish on the boundary.



From domain to matrix

Numerical
Approximation of
the Spectrum

Ebube Anozie

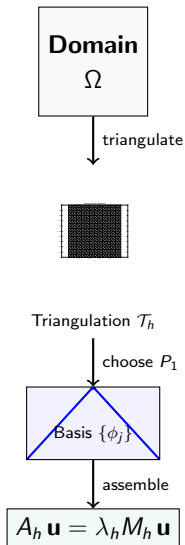
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion



Compactness: the engine behind discrete spectrum

Numerical
Approximation of
the Spectrum

Ebube Anozie

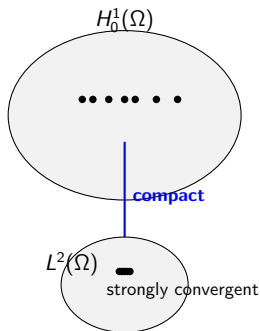
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion



Theorem 1 (Rellich-Kondrachov)

For a bounded Lipschitz domain $\Omega \subset \mathbb{R}^2$, the embedding $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact.

Consequence

Numerical
Approximation of
the Spectrum

Ebube Anozie

Introduction and
Motivation

Finite Element
Discretization

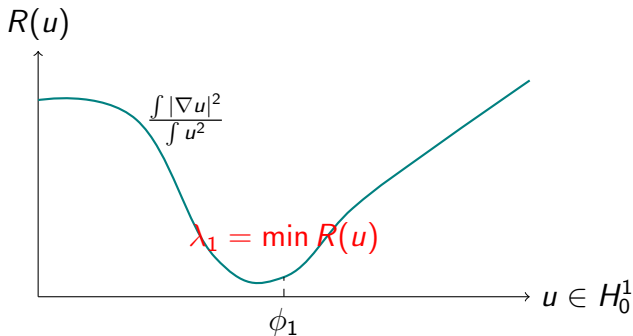
Convergence
Theory

Numerical
Experiment

Conclusion

The solution operator $T = (-\Delta)^{-1}$ is compact $\Rightarrow$ the Laplacian's spectrum is a discrete set
 $\lambda_1 < \lambda_2 \leq \dots \rightarrow \infty$.

Variational characterization



- Because of compact embedding, the minimum is attained by the first eigenfunction ϕ_1 .
- Discrete eigenvalues satisfy $\lambda_1 \leq \lambda_{1,h}$ (upper bound).

Convergence: $O(h^2)$ confirmed

Numerical Approximation of the Spectrum

Ebube Anozie

Introduction and Motivation

Finite Element Discretization

Convergence Theory

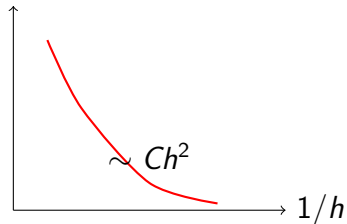
Numerical Experiment

Conclusion

Convergence of the principal eigenvalue $\lambda_{1,h}$ on the unit square.

h	Deg. of Freedom	$\lambda_{1,h}$	Error	Ratio
0.1000	121	20.2294	4.89e-01	---
0.0500	441	19.8611	1.22e-01	4.01
0.0250	1681	19.7697	3.04e-02	4.00
0.0125	6561	19.7468	7.61e-03	4.00
0.0063	25921	19.7411	1.90e-03	4.00

error



$$|\lambda_1 - \lambda_{1,h}| \leq Ch^2$$

A Chladni Experiment on the Unit Drum

Numerical
Approximation of
the Spectrum

Ebube Anozie

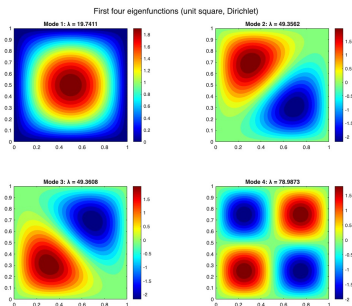
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion



- Mode 1: positive, fundamental tone
- Modes 2 & 3: degenerate, one nodal line
- Mode 4: cross nodal line, higher pitch

Orthogonality preserved

Numerical
Approximation of
the Spectrum

Ebube Anozie

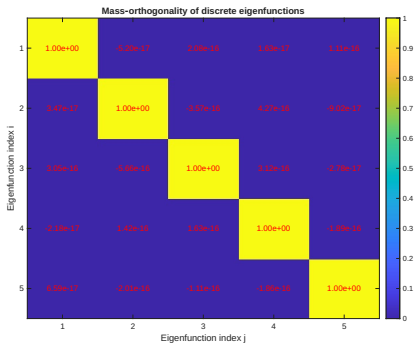
Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion



The mass matrix M_h preserves the self-adjointness of the continuous Laplacian.

Key Takeaways

- **Compactness is the engine.** The Rellich-Kondrachov theorem guarantees a discrete spectrum.
- **Finite elements turn PDEs into linear algebra.** Galerkin projection + P1 elements $\rightarrow$ sparse, symmetric matrix eigenproblem that we can solve efficiently.
- **$O(h^2)$ convergence is observed.** As the mesh is refined, the error shrinks quadratically – exactly as theory predicts.
- **The numerics respect the physics.** Degeneracies, nodal lines, and orthogonality all emerge naturally from the discretization, confirming that FEM preserves the essential spectral structure.

Future Work

Numerical
Approximation of
the Spectrum

Ebube Anozie

Introduction and
Motivation

Finite Element
Discretization

Convergence
Theory

Numerical
Experiment

Conclusion

- **Non-convex domains:** Corner singularities reduce convergence rate to $O(h^{2\alpha})$ with $\alpha < 1$.
- **Adaptive FEM** to recover optimal convergence.
- Extend to **non-coercive operators** (e.g., Dirac operator) and spectral pollution.

Thank you

Questions & Discussion

Ebube Anozie

`ebube.anozie@colorado.edu`